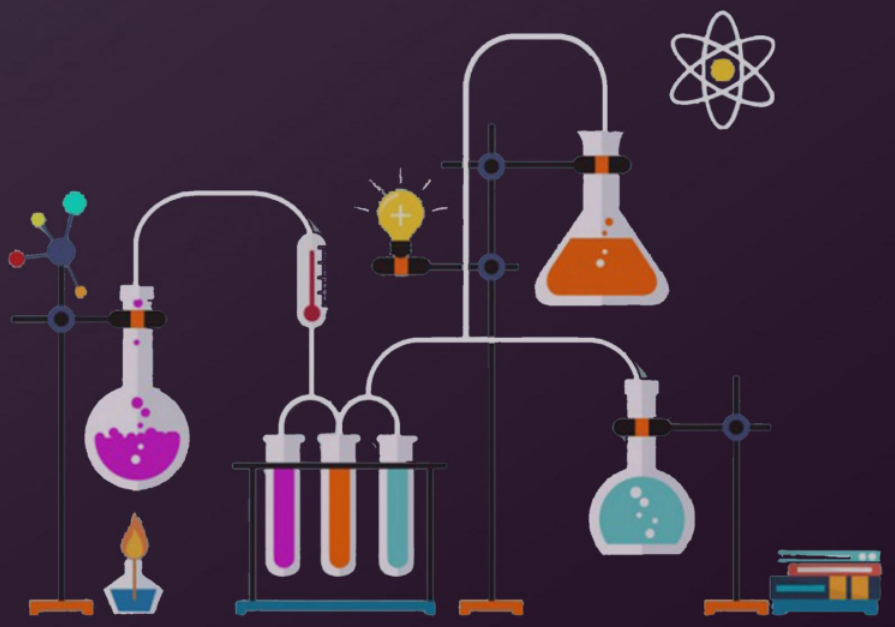
CHEMISTRY

Optical Isomerism



SHEET

eSaral.com

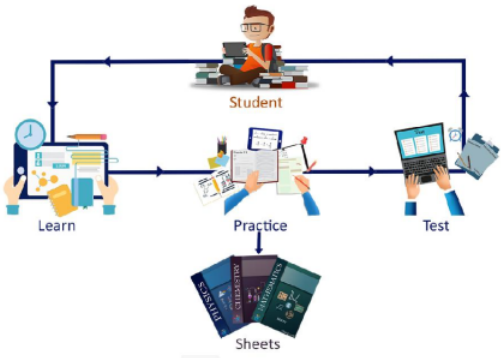
Contents

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2.	Homework Index	Page - iii
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Optical Isomerism		1	1	
2	Elements Of Symmetry (POS and COS)	1-7	2-6	2	1-4
3	2-D representation of 3-D molecule				
4	Enantiomers, Diastereomers And Meso Compounds	8-17	7, 13-14	3-9	5-8, 14-17, 20, 26
5	D,L and R,S configuration	18-21	8-11	10-14	9-10
6	Number of Stereoisomers, Resolvable and Non-resolvable Compounds		16, 17-19	15-17	24-25
7	Interconversions Of Projections				12-13
8	ACC	22	12, 15		11, 18-19, 21-23, 27

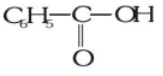
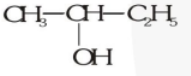
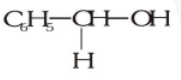
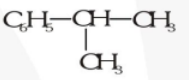
Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

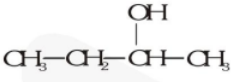
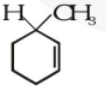
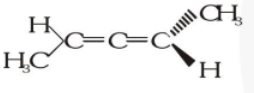
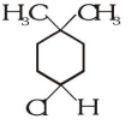
*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Elements of Symmetry (POS and COS)

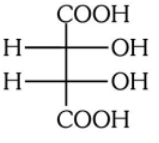
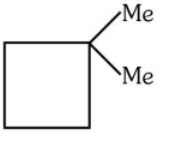
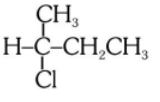
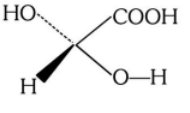
Q1. Which is optically active molecule :-

(A) 
(B) 
(C) 
(D) 

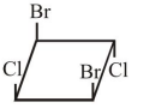
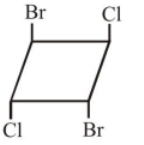
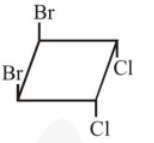
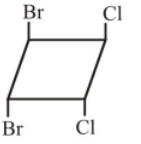
Q2. Which of the following compound is optically inactive:

(A) 
(B) 
(C) 
(D) 

Q3. Which of the following is optically active:-

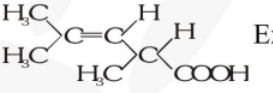
(A) 
(B) 
(C) 
(D) 

Q4. Which of the following structure is chiral -

(A) 
(B) 
(C) 
(D) 

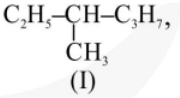
Q5. Number of POS present in CH_4.

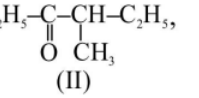
(A) 3 (B) 4
(C) 5 (D) 6

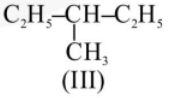
Q6.  Exhibits :-

(A) Tautomerism
(B) Optical isomerism
(C) Geometrical isomerism
(D) Geometrical and optical isomerism

Q7. Among the following structure I to III


(I)


(II)


(III)

It is true that :-

(A) All three are having chiral carbon
(B) Only I and II having chiral carbon
(C) Only II has chiral carbon
(D) Only I and III having chiral carbon

Enantiomers, Diastereomers And Meso Compounds

Q8. If a chiral compound has its specific rotation +40° then its mirror image :-

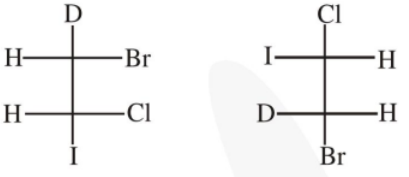
(A) will be laevorotatory

(B) will constitute racemic mixture if mixed with it in equal concentration

(C) will have specific rotation – 40°

(D) All the above are correct

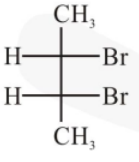
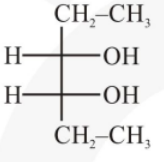
Q9. The two compounds given below are :-

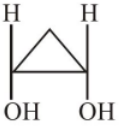


(A) enantiomers (B) identical

(C) optically inactive (D) diastereomers

Q10. Which of the following is a meso compound?

(A)  (B) 

(C)  (D) All of these

Q11. Which of the following compounds is a meso form of compound :-

(A)  (B) 

(C)  (D) 

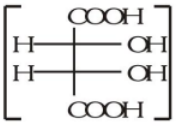
Q12. Meso-tartaric acid and d-tartaric acid are :-

(A) Positional isomers

(B) Enantiomers

(C) Diastereomers

(D) Racemic mixture

Q13. Meso–tartaric acid  is optically inactive due to the presence of :-

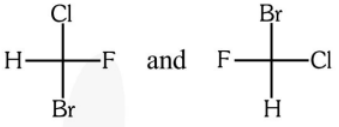
(A) Molecular symmetry

(B) Molecular asymmetry

(C) External compensation

(D) Two asymmetric carbon atoms

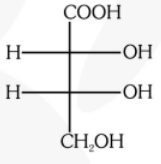
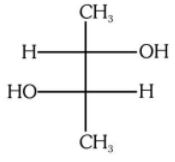
Q14. The following two compounds are

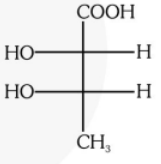
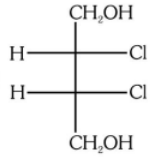


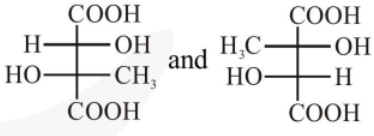
(A) Enantiomers (B) Diastereomers

(C) Identical (D) Epimers

Q15. Which one of the following is a meso–compound.

(A)  (B) 

(C)  (D) 

Q16.  are :-

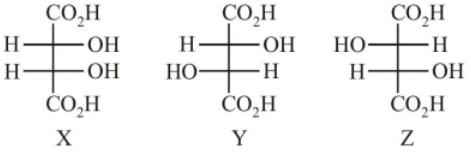
(A) Enantiomers

(B) Position isomers

(C) Geometrical isomers

(D) Identical

Q17. Which of the following option is correct :

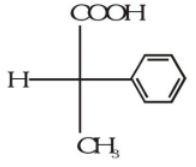


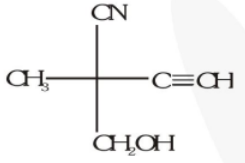
X Y Z

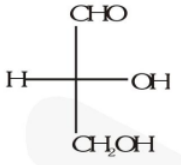
(A) X, Y is related as enantiomer
(B) X, Z is related as enantiomer
(C) Y, Z is related as enantiomer
(D) X, Y, Z all are optically active

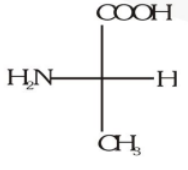
D, L And R, S Configuration

Q18. Identify R configuration :

(A) 

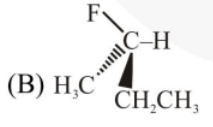
(B) 

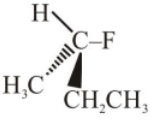
(C) 

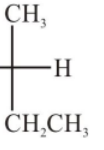
(D) 

Q19. The structure of (S)-2-fluorobutane is best represented by :-

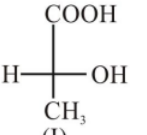
(A) 

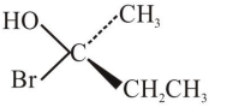
(B) 

(C) 

(D) 

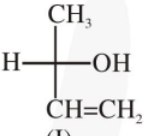
Q20. The correct configuration assigned for compounds (I) and (II) respectively are :-

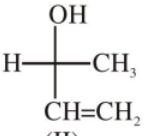

(I)

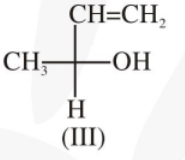

(II)

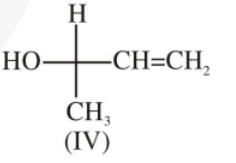
(A) R, R
(B) S, S
(C) R, S
(D) S, R

Q21. Which of the following combinations amongst the four Fischer projections represents the same absolute configurations?


(I)


(II)


(III)

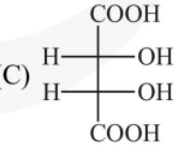

(IV)

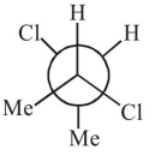
(A) (II) and (III)
(B) (I) and (IV)
(C) (II) and (IV)
(D) (III) and (IV)

ACC

Q22. Which of the following will not show optical isomerism.

(A) Cl–CH=C=C=CH–Cl
(B) Cl–CH=C=C=C=CH–Cl

(C) 

(D) 

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the given is correct specification for analysing optical activity.

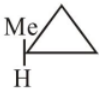
(A) 1 dm (Tube Length)

(B) 1 g/ml (Concentration)

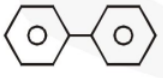
(C) 1 cm (Tube Length)

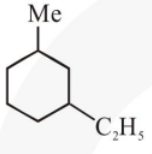
(D) Temp. (25°C)

Q2. Which of the following have POS.

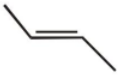
(A) 

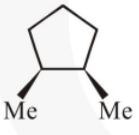
(B) 

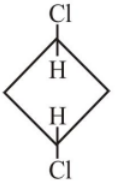
(C) 

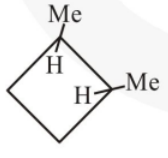
(D) 

Q3. Which of the following have COS -

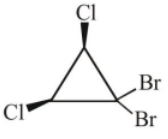
(A) 

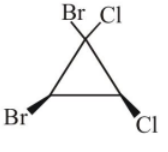
(B) 

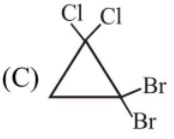
(C) 

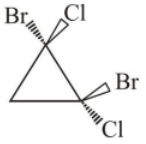
(D) 

Q4. Compound A having molecular formula $C_3H_2Cl_2Br_2$. A is optically active , A may be

(A) 

(B) 

(C) 

(D) 

Q5. Which of the following compounds is/ are optically inactive -

(A) $(CH_3)_2CHCH_2OH$

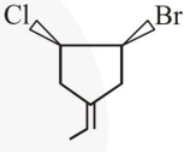
(B) CH_3CH_2OH

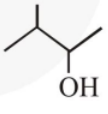
(C) CCl_2F_2

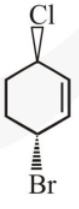
(D) $CH_3CHOHC_2H_5$

Q6. Which of the following compounds are optically active

(A) 

(B) 

(C) 

(D) 

Q7. Identify the compound which does not have any meso stereoisomer -

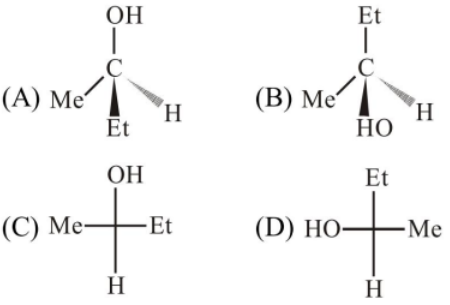
(A) 2-hydroxypropanoic acid

(B) 2,3-dichloropentane

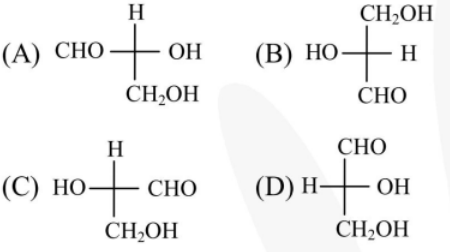
(C) 2-chlorobutane

(D) 2,3-dichlorobutane

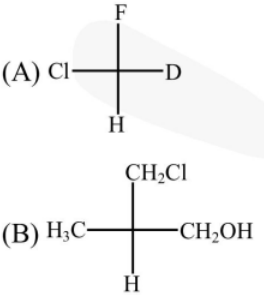
Q8. Which of the given represent R - 2 Butanol:-

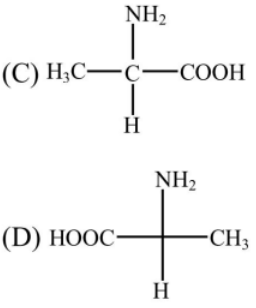


Q9. Identify the structure(s) which represent D-glyceraldehydes –

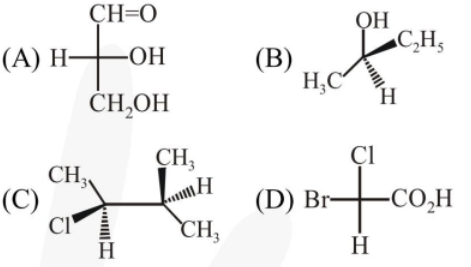


Q10. Identify, which of the following molecules is / are having 'R' configuration :

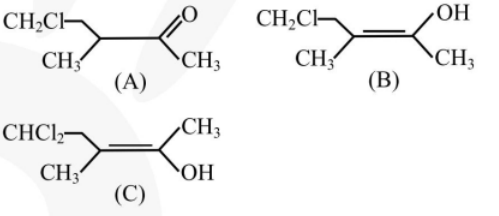




Q11. Which of the compound having R configuration.



Q12.



(A) A,C & A, B are tautomer
(B) A & B are tautomer
(C) A & C are resonating structure
(D) B & C are diastereomer

MATCHING LIST TYPE

Each question has TWO (02) ***matching lists: LIST I and LIST II****. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q13. Match List-I with List-II and select the correct answer using the codes given below the lists -

List-I	**List-II**
(a) Meso	(1) An equimolar mixture of enantiomers
(b) Enantiomers	(2) Stereoisomers that are not mirror images
(c) Diastereomers	(3) Non-super imposable mirror images
(d) Racemates	(4) An optically inactive compound whose molecule is achiral even though it contain chiral centres

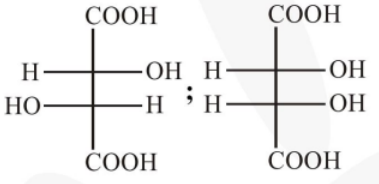
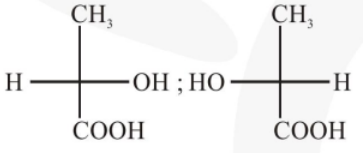
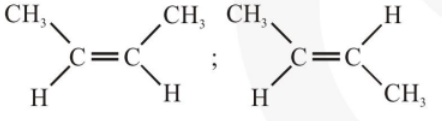
Codes :

	a	b	c	d
(A)	4	3	2	1
(B)	4	3	1	2
(C)	3	4	2	1
(D)	3	4	1	2

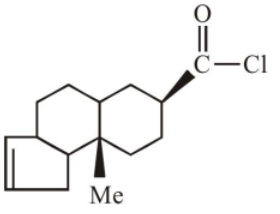
MATCH THE COLUMN TYPE

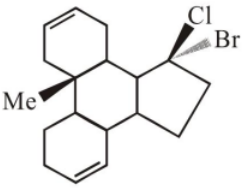
*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q14. Match the Column :-

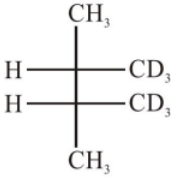
Column-I (Molecules)	**Column-II (Relationship)**
(A) 	(P) A pair of Metamers
(B) $CH_3OC_3H_7$; $C_2H_5OC_2H_5$	(Q) Tautomerism
(C) 	(R) A pair of geometrical isomers
(D) 	(S) A pair of diastereomers
(E) $CH_3CH_2CH_2-C(=O)-H$; $CH_3CH_2CH=CH-OH$	(T) A pair of optical isomers

Q15. Match Column :-

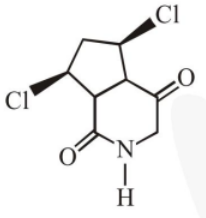
Column-I	**Column-II**
(A) 	(P) Compound have odd no. of Chiral Centres

(B) 

(Q) Compound have even no. of chiral centres

(C) 

(R) Optically active compound

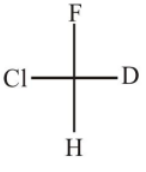
(D) 

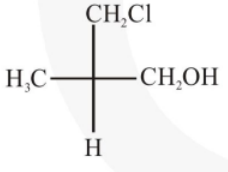
(S) Meso Compound

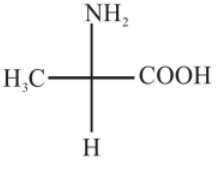
NUMERICAL TYPE

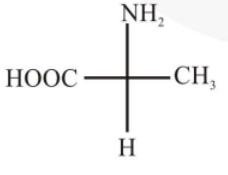
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

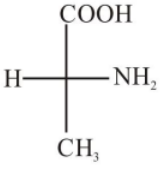
Q16. How many molecule have R - configuration :-



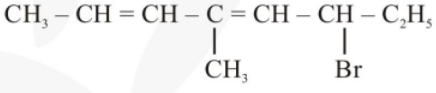




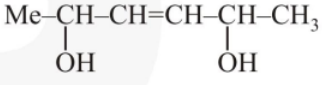




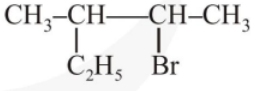
Q17. How many stereoisomer possible for



Q18. Identify total number of stereoisomers for the following compound :



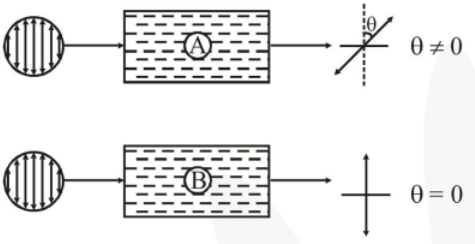
Q19. Identify total number of stereoisomers for the following compound :



SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1.



Which of the following statement is true about A and B :-

(A) A is optically active but B is optically inactive

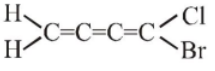
(B) A is optically inactive but B is optically active

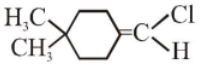
(C) Both A and B are optically active

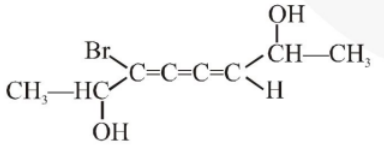
(D) Both A and B are optically inactive

Q2. The optically active compound is :

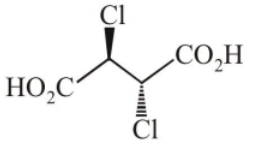
(A) 

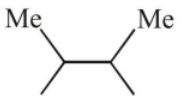
(B) 

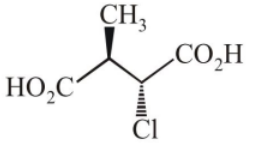
(C) 

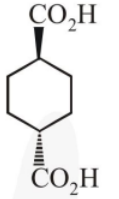
(D) 

Q3. Identify meso compound.

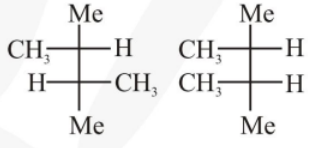
(A) 

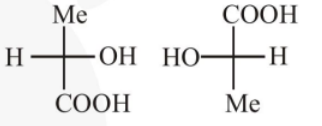
(B) 

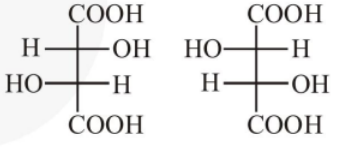
(C) 

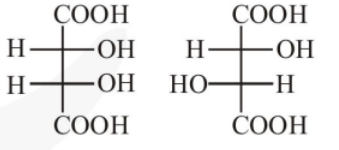
(D) 

Q4. Which of the following is a pair of Enantiomer

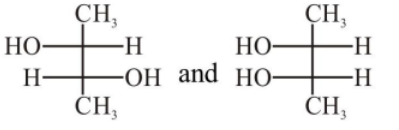
(A) 

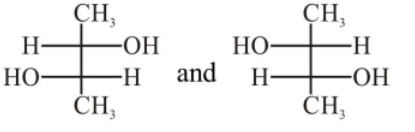
(B) 

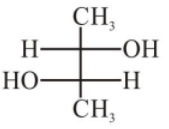
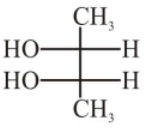
(C) 

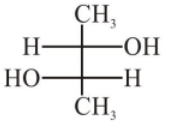
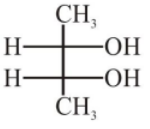
(D) 

Q5. Which of the following pairs of compounds are enantiomers :-

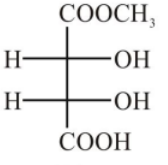
(A) 

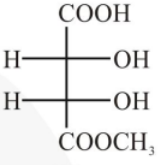
(B) 

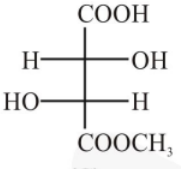
(C)  and 

(D)  and 

Q6. The correct statement about the compound A, B and C is :-


(A)


(B)


(C)

(A) A and B are identical

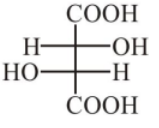
(B) A and B are diastereomers

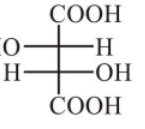
(C) A and C are enantiomers

(D) A and B are enantiomers

Q7. Which is incorrect about meso-tartaric acid:-

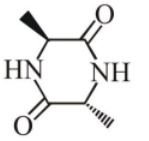
(A) It is optically inactive due to internal compensation

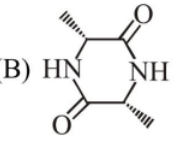
(B) It is enantiomer of 

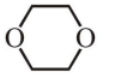
(C) It is diastereomer of 

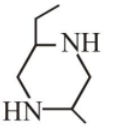
(D) It has two chiral carbon atoms

Q8. Which of the following is example of meso compound?

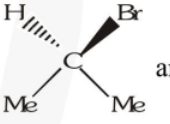
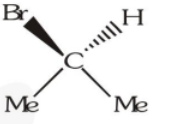
(A) 

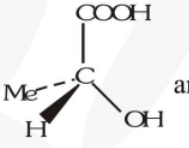
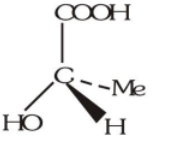
(B) 

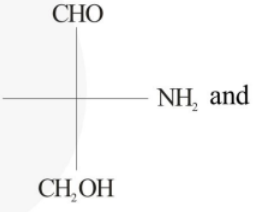
(C) 

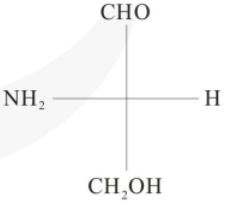
(D) 

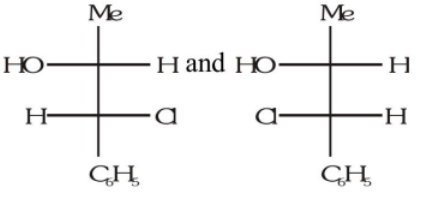
Q9. Which of the following pairs are diastereomers :

(A)  and 

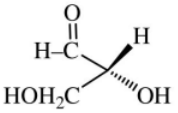
(B)  and 

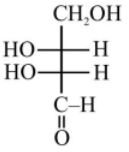
(C)  and

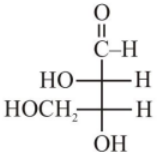


(D) 

Q10. Identify compounds having 'D' configuration ?

(A) 

(B) 

(C) 

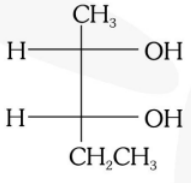
(D) All of these

Q11. The absolute configuration of



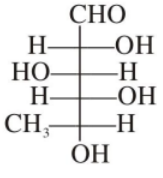
(A) S, R
(B) S, S
(C) R, R
(D) R, S

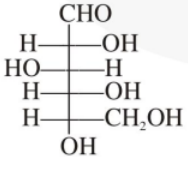
Q12. The absolute configuration at C_2 and C_3 of the compound given below is :-

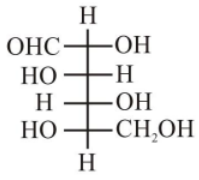


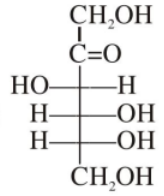
(A) (2R, 3S)
(B) (2S, 3R)
(C) (2S, 3S)
(D) (2R, 3R)

Q13. Which of the following is not D sugar :

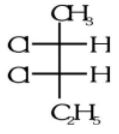
(A) 

(B) 

(C) 

(D) 

Q14. The absolute configuration of the compound is :

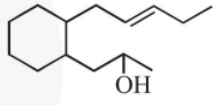


(A) 2S, 3R
(B) 2S, 3S
(C) 2R, 3S
(D) 2R, 3R

Q15. The number of optically active isomers of Glucose is :-

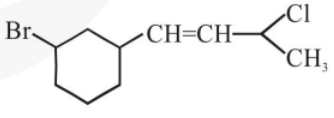
(A) 4
(B) 8
(C) 12
(D) 16

Q16. Number of optically active stereoisomers possible for the following compound :-



(A) 9
(B) 8
(C) 12
(D) 16

Q17. How many stereoisomer are possible for the following molecule ?

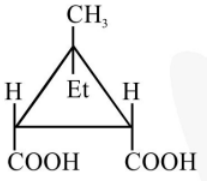


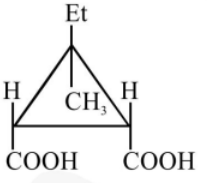
(A) 4
(B) 8
(C) 16
(D) 32

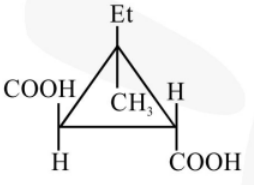
ONE OR MORE THAN ONE CORRECT TYPE

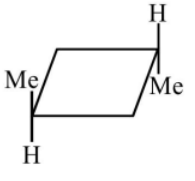
Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following are optically inactive:-

(A) 

(B) 

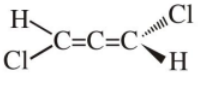
(C) 

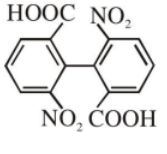
(D) 

Q2. Which of the following compounds are optically active ?

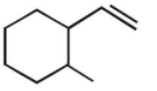
(A) $CH_3.CHOH.CH_2.CH_3$

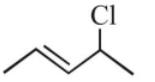
(B) $H_2C{=}CH.CH_2.CH{=}CH_2$

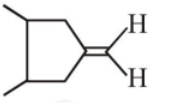
(C) 

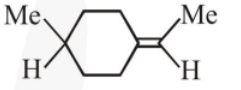
(D) 

Q3. Which of the following can show geometrical and optical isomerism .

(A) 

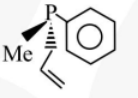
(B) 

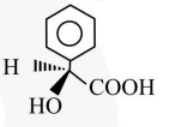
(C) 

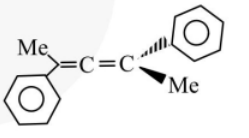
(D) 

Q4. Which of following is optically active -

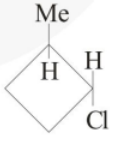
(A) $\mathbf{CH_3—CH_2—Cl}$

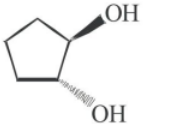
(B) 

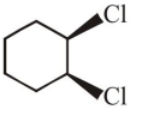
(C) 

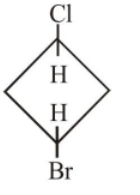
(D) 

Q5. Which of the following is Meso :-

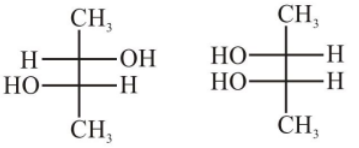
(A) 

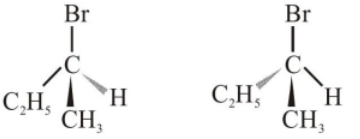
(B) 

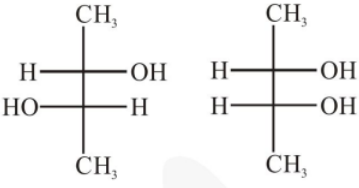
(C) 

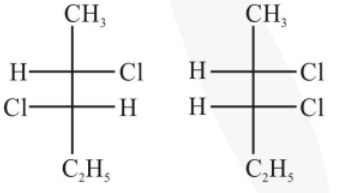
(D) 

Q6. Which are not enantiomers -

(A) 

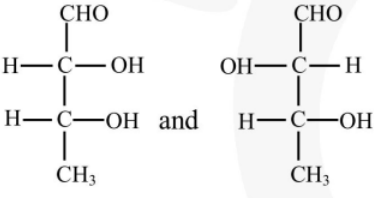
(B) 

(C) 

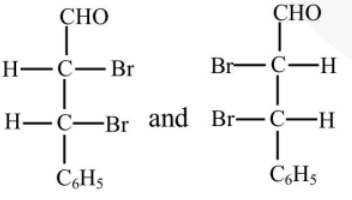
(D) 

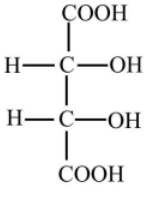
Q7. Which one of the following pairs is not correctly matched ?

(A) enantiomers :

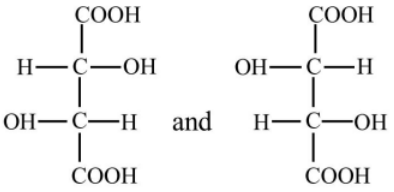


(B) Diastereomers :

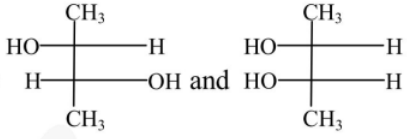


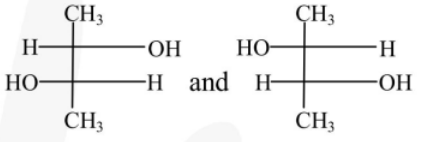
(C) meso form : 

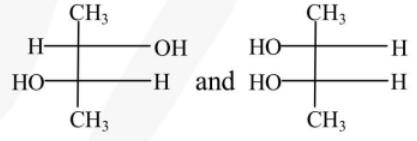
(D) Identical :

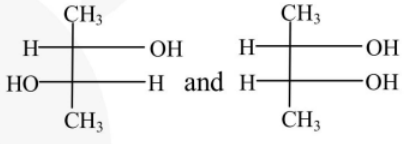


Q8. Which of the following pairs of compounds are not enantiomers ?

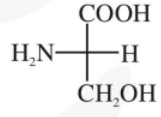
(A) 

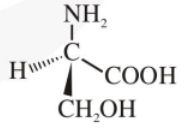
(B) 

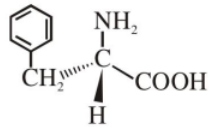
(C) 

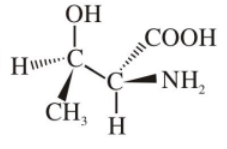
(D) 

Q9. Which of the following are correct representation of L-amino acids :

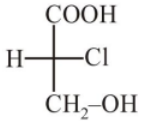
(A) 

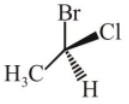
(B) 

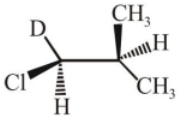
(C) 

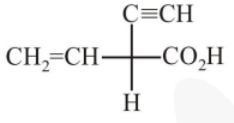
(D) 

Q10. Which of the compound having S configuration.

(A) 

(B) 

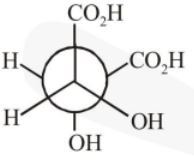
(C) 

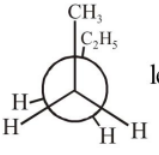
(D) 

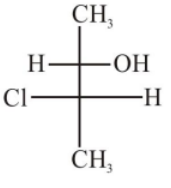
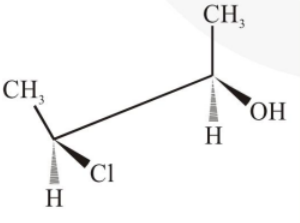
Q11. Which of the following options is in correct?

(A) Racemic mixture is optically inactive because of internal compensation

(B) Meso compound is optically inactive because of external compensation

(C)  is meso compound

(D)  least stable conformational form of n-hexane

Q12.  

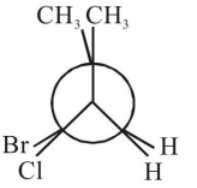
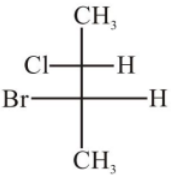
Above examples are not -

(A) Identical

(B) Enantiomers

(C) Diastereomers

(D) Non superimposable mirror images

Q13.  

By considering the above structure, tick the correct statement.

(A) Both are enantiomers

(B) Both are identical

(C) Both are diastreomers

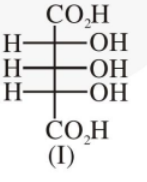
(D) Fisher can be drawn only for eclipsed form

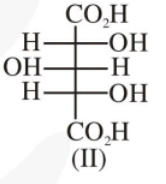
PARAGRAPH TYPE

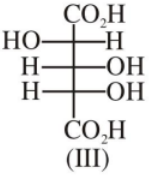
*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

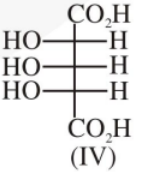
Paragraph for Q.14 to 15

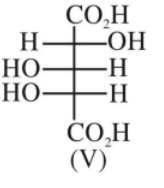
Observe the given compounds and answer the following questions.


(I)


(II)


(III)


(IV)


(V)

Q14. Which of the above formulae represent identical compounds ?

(A) I and II (B) I and IV

(C) II and IV (D) III and IV

Q15. Which of the above compounds are enantiomers ?

(A) II and III

(B) III and IV

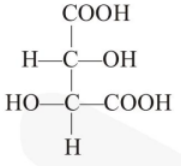
(C) III and V

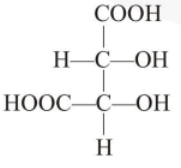
(D) I and V

Paragraph for Q.16 to 17

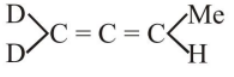
Optically active compound may or may not have chiral centre, optical activity of compounds are related with dissymmetric nature. Compound is dissymmetric if it has neither plane of symmetry nor centre of symmetry nor alternating axis of symmetry. Optical isomers may be optically active or may be optically inactive. Optically active compound show enontiomerism.

Q16. Identify optically inactive compound among following :-

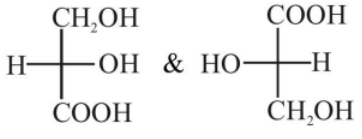
(A) 

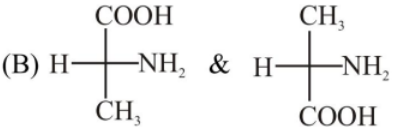
(B) 

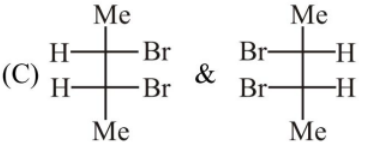
(C) 

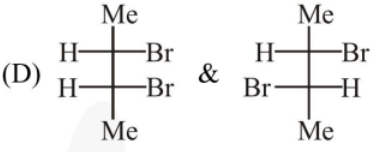
(D) 

Q17. Select correct enantiomeric pair among following :-

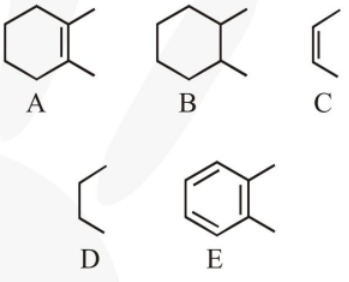
(A) 

(B) 

(C) 

(D) 

Paragraph for Q.No. 18 to 19



Q18. Compound showing geometrical isomerism

(A) A only (B) B only

(C) B & C (D) C & D

Q19. Compound showing optical isomerism

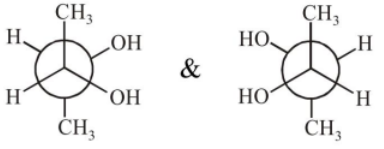
(A) A only (B) B only

(C) B & C (D) D & E

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching For each question, choose the option corresponding to the correct matching.*

Q20.

List-I (Compounds / Orientations)		**List-I (Relation)**
(P) 		(1) Position isomers
(Q) 		(2) Diastereomers
(R) 		(3) Geometrical isomers
(S) 		(4) Enantiomers

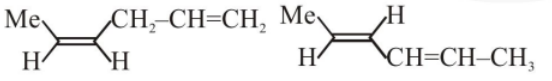
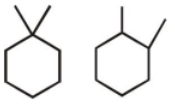
Code :

	P	Q	R	S
(A)	2,3	2,3	4	4
(B)	2,3	2,3	4	1
(C)	2,3	2,3	2	4
(D)	2,3	2,3	2	2

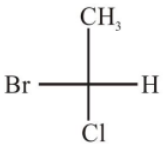
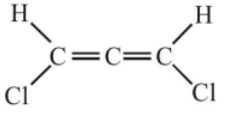
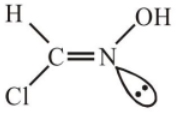
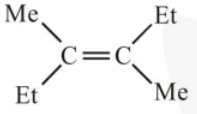
MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

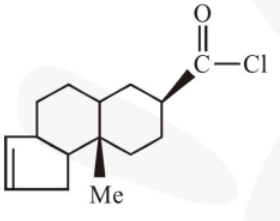
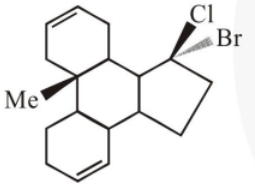
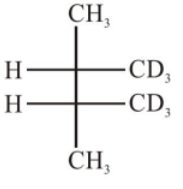
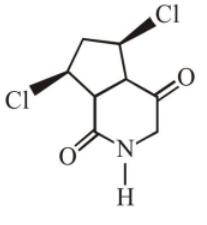
Q21.

Column–I	**Column–II**
(A) 	(P) Geometrical isomers
(B) 	(Q) At least one of the compound can show Geometrical Isomerism
(C) 	(R) At least one of the compound can show Optical Isomerism
(D) $CH_2=CH-CH=CH-CH_3CH_3-CH=C=CH-CH_3$	(S) Positional isomers

Q22. Match Column (I) and (II) :-

Column–I	Column–II
(A) 	(P) Optically active compound
(B) 	(Q) Compound contains chiral centre
(C) 	(R) Compound contains stereo center
(D) 	(S) Compound contains plane of symmetry

Q23. Match Column :-

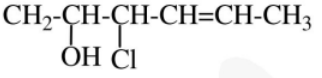
Column-I	Column-II
(A) 	(P) Compound have odd no. of Chiral Centres
(B) 	(Q) Compound have even no. of chiral centres
(C) 	(R) Optically active compound
(D) 	(S) Meso Compound

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q24. How many planes are possible in benzene to make plane of symmetry?

Q25. Identify total number of stereoisomers for the following compound :



ASSERTION AND REASON

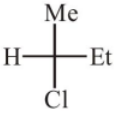
(A) Statement–1 is true, Statement–2 is true; Statement–2 is not the correct explanation of Statement–1.

(B) Statement–1 is true, Statement–2 is true; Statement–2 is the correct explanation of Statement–1.

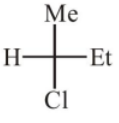
(C) Statement–1 is true, Statement–2 is false.

(D) Statement–1 is false, Statement–2 is true.

Q26. **Statement–1 :**

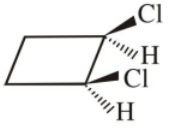
 is a chiral resolvable molecule.

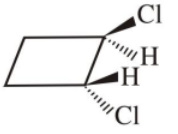
Statement–2 :

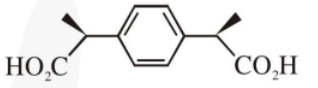
 is non-superimposable on its mirror image.

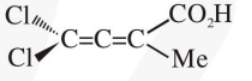
SUBJECTIVE TYPE

Q27. With reasons, state whether each of the following compounds I to VIII is chiral

(I) 

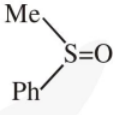
(II) 

(III) 

(IV) 

(V) 

(VI) 

(VIII) 

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
B	D	C	D	D	B	B	D	A	D
11	12	13	14	15	16	17	18	19	20
C	C	A	A	D	D	C	C	C	A
21	22								
C	A								

Ex - 1A

1. A, B, D **2.** A, B, C **3.** A, C **4.** B, D **5.** ABC

6. A, B, C, D **7.** A, B, C **8.** A,C,D **9.** B,C,D **10.** A,B,C

11. A,B,D **12.** B **13.** A

14. A → S, T; B → P; C → T; D → R, S; E → Q

15. A → Q, R; B → P, R; C → S, Q; D → Q, R

16. 4 **17.** 8 **18.** 6 **19.** 4

Ex-2									
1	2	3	4	5	6	7	8	9	10
A	D	A	C	B	D	B	A	D	D
11	12	13	14	15	16	17			
C	B	B	C	D	D	C			

Ex - 2A

1. A, B, D **2.** A, C, D **3.** A, B, C **4.** B, C, D **5.** C

6. A, C, D **7.** A, B, D **8.** A, C, D **9.** A, C, D **10.** C, D

11. A, B, D **12.** A, B, D **13.** B, D **14.** B **15.** C

16. A, C, D **17.** B **18.** C **19.** B **20.** B

21. (A)→P,Q ; (B)→Q,S ; (C)→Q,R,S ; (D)→Q,R,S

22. A → P, Q, R; B → P, R; C → R, S; D → R, S

23. A → Q, R; B → P, R; C → S, Q; D → Q, R

24. 7 **25.** 8 **26.** B

27. Achiral : I, III , IV ; Chiral : II,V, VI, VII